# Cycle Sequencing



#### SUBMISSION — 13 µL IN A 1.5 ML TUBE

**3** G00101 sequencing oligo (2.66 µM)
**10** miniprep DNA

**13 µL total**

Our sequencing oligos are standardised to **2.66 µM**, so **3 µL is exactly one reaction's worth**. The other 10 µL is DNA and water.

#### TUBES AND LABELS

- **1.5 mL tubes. Not PCR tubes.**
- Label the **top** of the tube.
- The label is **the exact name you put on the submission form** — not an abbreviation, not a name that only makes sense to you. The facility matches tube to form by that string.

#### HOW MUCH DNA — BY COPY NUMBER

- **Medium copy** — pBR322, pAC, and **pP6**: **10 µL** miniprep.
- **High copy** — pUC plasmids: about **4 µL** miniprep, made up to 10 µL with water.
- **Low copy** — pSC101, BACs: **PCR the region first** and sequence the PCR product.

#### USE THE DGTP PROTOCOL

Not the standard chemistry. Standard chemistry **dies inside a hairpin** — a terminator, or any strong secondary structure, stops the read dead. This cost Tlib2 an entire sequencing run.

Say **dGTP** on the submission form. Results come back in **1–2 days**.

#### WHAT YOU GET

One primer, so extension is **linear**, not exponential. The read starts **20–50 bp downstream** of the primer and gives **400–1000 bp** of usable sequence. Pick a primer **upstream** of what you want to see — for pP6, **G00101**.

#### SEQUENCE EVERYTHING YOU PICKED

Including the boring-looking ones. The clones in the **middle of the range** are the informative ones. Tlib2 sequenced only its brightest and left its main question unanswered for three years.

#### CHECK THE READ

A `.txt` of base calls (the *read*) and an `.ab1` trace. Open both in **ApE**; `ctrl-K` annotates features.

1. Is it **clean**? Length with no Ns: 100 bp poor, 800 good, 1000 great.
2. Architecture **BseRI** → **promoter** → **BseRI**. Exactly **two** BseRI sites.
3. **T4 terminator** present; promoter not duplicated, reversed or truncated.
4. Align to `pP6.seq` — **Tools** → **Align with another sequence...**
5. Search for the motif below. Clean read + motif present = **usable**.

```
GAGGAGTCCTGGGTTCNNNNTTGACANNNNNNNNNNNNNNNNNTATAA
TNNNNNNANNNNGTTAGTATTTCTCCTC
```

#### RECORD

**exp · clone_id · student_name · read_name · date_sequenced · canonical · usable · cassette · notes**

- **canonical** = matches `pP6.seq` across the good-quality region. **usable** = contains the motif. Different questions.
- Expect artifacts — duplications, deletions, recombinations, and some plain parent plasmid. Record them as artifacts and exclude them.